



# Investigation on datasets toward intelligent intrusion detection systems for Intra and inter-UAVs communication systems

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## ABSTRACT

UAVs, commonly known as drones, are increasingly utilized in various fields such as military operations, surveillance, agriculture, and delivery services. Given their expanding roles, several critical reasons underline the necessity for security and trustworthiness including sensitive operations, and Public Safety and Privacy. Effective and secure communication channels are fundamental to the reliability and efficiency of UAVs. This involves protecting data transmission from various threats and ensuring that the communication protocols are robust against attacks. The security of communication systems directly impacts the overall trustworthiness of UAV operations, making it a critical area of focus for researchers and developers in this field. Intrusion Detection Systems (IDS) are vital components of a resilient architecture, essential for ensuring the security of communication systems. These systems function as diligent sentinels, continuously monitoring network traffic for signs of malicious behavior or rule violations. Additionally, IDS plays a crucial role in protecting communication networks by promptly identifying and responding to potential threats. Integrating IDS with other security measures enables organizations to significantly enhance their overall security posture. Accordingly, several recent studies have proposed advanced approaches using artificial intelligence to enhance the security and trustworthiness of UAV, as well as to mitigate cyberattacks in real time. However, the efficiency of the proposed AI-based approaches relies on the learning phase, which is strongly correlated with the quality of the used datasets. In this context, the objective of this article is to offer a thorough investigation concerning this topic. Certainly, the development and testing of effective AI-based IDS for UAVs require access to a diverse range of datasets that precisely capture the various security challenges and potential attack scenarios. In light of this necessity, the current study undertakes a comprehensive examination of UAV communication systems and their associated networking architectures, both centralized and decentralized. This study specifically focused on Intra and Inter UAV communication systems and utilized relevant IDS datasets. Additionally, it delves into the realm of open datasets that are pertinent to intelligent intrusion detection systems. More specifically, this paper introduces a novel taxonomy designed to categorize datasets relevant to IDS within the UAV context. Furthermore, it furnishes a guide along with recommendations for the selection of appropriate datasets based on predetermined scenarios.

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## 1. Introduction

### 1.1. General context

In recent years, the proliferation of Unmanned Aerial Vehicles (UAVs) (Mohammed et al., 2024b), commonly referred to as drones, has opened new horizons in various sectors and has found applications spanning diverse industries, including military operations (Wang et al., 2020), mobile relay nodes within cellular networks disaster management, smart agriculture (Jasim et al., 2023; Qu et al., 2022), target tracking (Hentati et al., 2022), and commercial deliveries, making their presence ubiquitous. UAVs are encompassed within the broader framework of Unmanned Aerial Systems (UAS).

As UAV technology advances and their applications diversify, the vulnerabilities and threats associated with these systems become increasingly apparent, with cyberattacks emerging as a critical concern (Hassija et al., 2021). The escalating integration of drones in various sectors has also witnessed a corresponding rise in cyber-attacks targeting UAVs, the communication links associated with the UAV

systems, the deployed protocols, and the applications embedded within UAVs (Tlili F. et al., 2022).

As UAV technology continues to evolve, various architectural paradigms and communication frameworks have emerged to support the growing complexity and sophistication of UAV operations (Hentati and Fourati, 2020). Indeed, specific applications demand the utilization of a single UAV controlled remotely through a Ground Control Station (GCS) (Meng et al., 2023), commonly referred to as a mono-UAV system (Bank et al., 2022). Conversely, there are other types of applications that require the deployment of multiple drones operating in groups or networks to execute tasks and sophisticated missions collaboratively.

**Sensitive Operations:** UAVs often perform tasks involving sensitive information or operations that can have significant repercussions if compromised. For instance, in military and surveillance applications (Zhang et al., 2020; Mishra et al., 2022), unauthorized access to UAVs can lead to breaches of national security.

**Autonomous Functions:** Many UAVs operate autonomously or semi-autonomously, relying on complex algorithms and communication systems (Al-Wathinani et al., 2023). Security breaches could lead to loss

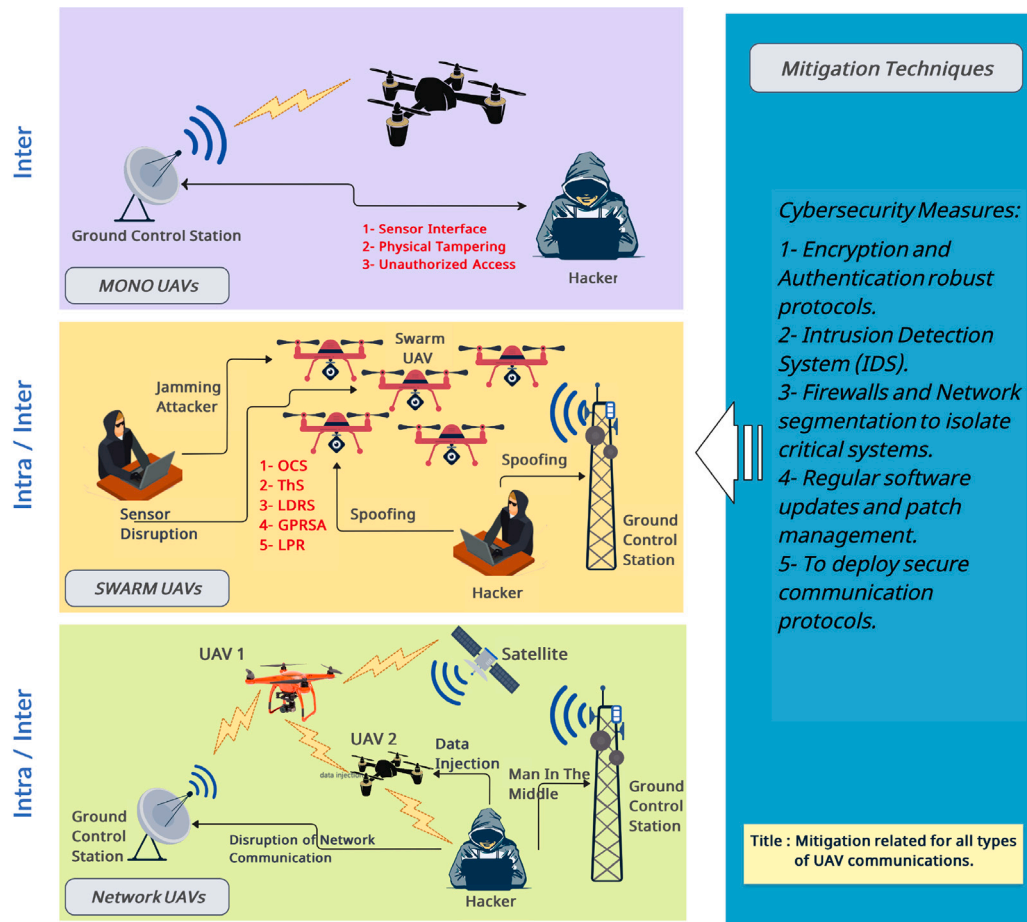


Fig. 1. UAV system communication types with mitigation techniques.

of control or malicious interference, potentially causing accidents or unauthorized actions.

**Communication Systems:** UAVs depend heavily on communication systems to interact with control stations, other UAVs, and possibly satellites (Abdelhafidh et al., 2022). Ensuring the integrity and security of these communication channels is crucial to prevent eavesdropping, jamming, spoofing, or other cyber threats that can disrupt their operations.

**Public Safety and Privacy:** As UAVs become more prevalent in civilian applications (Krichen et al., 2018), ensuring their security is vital to protect public safety and privacy. Unauthorized access or control over UAVs can lead to privacy invasions or even physical harm.

Within mono-UAV systems, UAV are remotely controlled via diverse communication networks that are susceptible to a range of cyber threats. as in Fig. 1: These threats encompass various attack vectors, including:

- **Communication interception:** Malicious actors may intercept communication between the UAV and its control station, gaining unauthorized access to sensitive flight instructions and sensor data. This interception jeopardizes the confidentiality and integrity of UAV operations.
- **Spoofing and Jamming:** Adversaries can manipulate GPS signals, leading UAVs off-course or causing them to crash (Chaari et al., 2020b,a). Additionally, jamming techniques disrupt communication links, severing the connection between the UAV and its operator, thereby rendering the UAV non-responsive.
- **Malware Injection:** Cyber-attackers can inject malware into UAV systems through compromised software updates or malicious payloads, enabling them to gain unauthorized control over the UAV's operations.

Swarm UAVs (Khelifi and Butun, 2022) that are characterized by their collaborative behavior and synchronization aspects, introduce new dimensions to cyber-security challenges such as:

- **Swarm Coordination Interference:** Cyber-attackers may exploit vulnerabilities in the communication protocols that facilitate swarm coordination (Javaid et al., 2023). This interference disrupts the synchronization and collaboration among swarm members, hampering their collective performance.
- **Decoy and Distraction Tactics:** Adversaries might manipulate the behavior of individual UAVs within a swarm, creating decoys or distractions to confuse or divert the swarm from its intended mission.

Networked UAVs, where multiple UAVs form interconnected systems, amplify the potential impact of cyberattacks such as the propagation of malicious code, and the execution of synergistic attacks.

- **Propagation of Malicious Code:** A cyberattack on one UAV within a networked system can facilitate the propagation of malicious code to other UAVs through shared communication links, potentially leading to a cascade of compromised UAVs.
- **Synergistic Attacks:** Networked UAVs rely on coordinated actions for complex missions. Attackers could exploit vulnerabilities in the communication infrastructure to execute synergistic attacks that disrupt the execution of collaborative tasks (Liu, 2023).

The growth of UAV technology has led to a new frontier in aerial operations. However, this progress comes with an increasingly critical need to address the cybersecurity challenges associated with UAV systems and secure the communication channels that enable the exchange

of data and commands among UAVs, particularly in the context of Swarm UAVs and Networked UAVs. As UAVs become more interconnected, the potential attack surface widens, necessitating robust intrusion detection mechanisms capable of identifying and mitigating diverse cyber threats. Intelligent Intrusion Detection Systems (Nuaimi et al., 2023; Hentati and Fourati, 2021) equipped with advanced algorithms and machine learning capabilities offer a promising solution to this escalating challenge.

Fig. 1 aims to visually convey the diversity of surface attack scenarios in single UAV, swarm UAVs, and networked UAVs contexts, emphasizing the critical importance of securing UAV systems against these threats.

### 1.2. Motivation and problem statement

Achieving robust security in UAV communication systems presents multifaceted challenges. One of the pivotal challenges lies in the development and deployment of effective and intelligent Intrusion Detection Systems tailored to the unique characteristics of UAV networks. In particular, intelligent IDS that harness the power of advanced algorithms and machine learning techniques are positioned to provide enhanced threat detection capabilities. Yet, the success of these intelligent IDS hinges on their training and validation datasets, which capture the complexities of real-world intrusion scenarios.

In this context, this research stems from the recognized gap in understanding the significance of appropriate datasets in advancing the field of Intelligent Intrusion Detection Systems for both Intra and Inter UAVs communication systems. The lack of comprehensive investigation into the selection, generation, and utilization of datasets tailored to UAV communication environments impedes the development of IDS that can effectively combat emerging cyber threats.

To address this problem, a comprehensive exploration of dataset diversity and realism becomes imperative. This entails investigating the diversity and realism inherent in datasets designated for training and testing intrusion detection algorithms within UAV communication contexts. This exploration aims to verify whether these datasets adequately account for real-world scenarios and to scrutinize the precision with which attack vectors are portrayed. Ensuring that these datasets accurately represent the multifaceted attack landscape guarantees the reliability and effectiveness of the IDS.

Furthermore, various factors influence the efficacy and robustness of intrusion detection algorithms, encompassing: (i) the development of methodologies for generating adversarial attacks within UAV communication datasets; and (ii) the evaluation of network dynamics, which involve UAV mobility, evolving communication topologies, and fluctuating congestion levels within the datasets.

### 1.3. Contributions of this study

This research conducts a thorough examination of the crucial role that datasets play in improving the development of Intelligent Intrusion Detection Systems designed particularly for communication systems of both Intra and Inter Unmanned Aerial Vehicles.

- The main significance of this study is to emphasize the crucial role of carefully choosing, organizing, and using suitable datasets for training and validating Intrusion Detection Systems in the field of communication between Unmanned Aerial Vehicles, both within and between UAVs. This is essential for ensuring the security of UAV missions and applications in a rapidly changing technological environment.
- This study aims to close the gap between the growing area of UAV communication security and the fundamental importance of realistic datasets by tackling this important unresolved problem.
- Current work serves as a crucial asset for scholars and programmers to contribute toward the advancement of intrusion detection systems that are more robust and efficient.

- This resource provides essential information and instructions on effectively integrating datasets into the creation, assessment, and enhancement of Intrusion Detection Systems for both intra/inter Unmanned Aerial Vehicle networks.

### 1.4. The structure of this paper

The remainder of this article is structured as follows: The second section reviews the existing surveys and investigations related to IDS within UAV communication environments. It highlights the studied datasets for securing intelligently UAV communication systems, and pinpoints gaps in the research landscape. The third section, overviews UAV communication systems, IDS architectures, and integration within UAV communication systems. The fourth section explores the essential attributes and characteristics of datasets that contribute to the effectiveness of intrusion detection algorithms. Discusses the factors that define a realistic and representative dataset for UAV communication scenarios, and presents our proposed taxonomy and classification of datasets used for intelligent approaches to secure UAV communication systems. The fifth section provides an extensive survey of existing datasets suitable for training and evaluating intrusion detection algorithms in the context of UAV communication. Examines the strengths and limitations of each dataset in terms of diversity, scalability, and real-world relevance. The sixth section provides guidelines for researchers and practitioners to judiciously integrate open datasets into the design and evaluation of intelligent intrusion detection systems for intra- and inter-UAV communication. Besides that, it delineates an extensive analysis and evaluation of various datasets tailored specifically for IDSs within UAV communication systems, elucidating their suitability, strengths, and limitations in this specialized domain. The last section of this paper summarizes the key findings and contributions of the study. Identifies avenues for future research, emphasizing the evolving nature of UAV communication security and the continuous need for relevant datasets. The roadmap of this work is shown in Fig. 2.

### 1.5. Acronyms

provides an extensive description of the abbreviations used in the present study.

## 2. Existing surveys and studies related to UAV-IDS datasets

Despite the growing demand for datasets in UAV research, only a few surveys and studies comprehensively review and compare the existing datasets dedicated to intrusion detection. Differentiating from this paper that focuses on reviewing datasets for IDS within UAVs, authors in Ajibuwa et al. (2023) focus on establishing objective evaluation metrics and assessing the feasibility of AI/ML-based IDS/MDS in networked autonomous systems, emphasizing the safety-critical aspects and practical adoption in AV/UAV networks. A lack of implementation of ideas in practical applications, despite substantial study. It was discovered that the assessment measures employed in the literature were insufficient for gauging the viability of these solutions. This study examines several aspects of autonomous cars, UAV networks, and V2X communication systems, including their current architecture and potential dangers. It then suggests ways to improve these systems by leveraging methods from other machine learning fields. The transportation industry is increasingly reliant on digital technologies like artificial intelligence and machine learning, making it vulnerable to cyber threats. The Internet of Everything (IoE) has increased the need for security and privacy in vehicles, with automotive electronics accounting for (50%) of total vehicle expenditure by 2030.

This study (Karopoulos et al., 2022) aims to create a comprehensive classification system for In-Vehicle Network Intrusion Detection Systems (IVN-IDS) for future research. This Comprehensive Review (Kheddar et al., 2023) explores the use of AI in Industrial Control



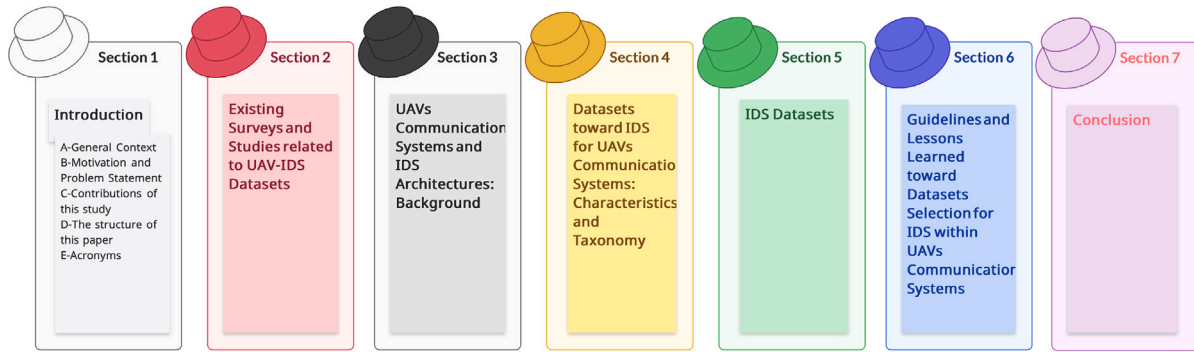


Fig. 2. Roadmap of this work.

Table 1

Acronym.

| Acronym | Description                                    |
|---------|------------------------------------------------|
| UAVs    | Unmanned Aerial Vehicles                       |
| UAS     | Unmanned Aerial System                         |
| IDS     | Intrusion Detection System                     |
| IDSs    | Intrusion Detection Systems                    |
| IoV     | Internet of Vehicles                           |
| IoT     | Internet of Things                             |
| IoD     | Internet of Drone                              |
| GCS     | Ground Control Station                         |
| NIDS    | Network Intrusion Detection System             |
| HIDS    | Hybrid Intrusion Detection System              |
| DDoS    | Distributed Denial of Service                  |
| LEACH   | Low Energy Aware Cluster Hierarchy             |
| FCL -   | Federated Continuous Learning -                |
| SBLs    | Stacked Broad Learning System                  |
| LGBM    | Light Gradient Boosting Machine                |
| IVN-IDS | In-Vehicle Network Intrusion Detection Systems |
| ALFA    | AirLab Failure and Anomaly                     |
| LSTM    | Long short-term memory                         |
| MAVLink | Micro Air Vehicle Link                         |
| MIT     | Massachusetts Institute of Technology          |
| GRU     | Gated Recurrent Unit                           |
| SGD     | Stochastic gradient descent                    |
| DTN     | Delay Tolerant Network                         |
| GAN     | Generative Adversarial Network                 |
| SSO     | Sparrow Search Optimization                    |
| CTF     | Capture the Flag                               |
| MITM    | Man-in-the-Middle                              |
| LOF     | Local Outlier Factor                           |
| BWO     | Black Widow Optimization                       |
| SSO     | Sparrow Search Optimization                    |
| ANN     | Artificial Neural Networks                     |
| CNN     | Convolutional Neural Networks                  |
| SVM     | Support Vector Machine                         |
| MLP     | Multilayer Perceptron                          |
| DNN     | Deep Neural Network                            |
| V2V     | Vehicle-to-Vehicle                             |
| KNN     | k-Nearest Neighbor                             |
| ML      | Machine Learning                               |
| FL      | Federated Learning                             |
| DL      | Deep Learning                                  |
| DT      | Decision Tree                                  |
| KM      | K-Means clustering                             |
| LR      | Logistic Regression                            |
| NB      | Naive Bayesian                                 |
| AD      | Anomaly Detection                              |
| AE      | Autoencoder                                    |

Systems using Deep Learning models to improve security and detect abnormal behavior. It highlights the need for IoT security measures and techniques to mitigate individual detection risks. Future trends include increased model resilience, transparency, data analysis, unsupervised learning integration, and FTL approaches. This systematic survey (Momand et al., 2023) examines intrusion detection systems

with regard to their design, detection methodologies, decision-making, and localization. The text focuses on machine learning-based intrusion detection system techniques and deep learning models. Nevertheless, concerns regarding efficiency and performance are recognized. The current body of research mostly concentrates on a limited number of datasets, rendering networks more vulnerable to intrusions. Updating datasets is essential to meet contemporary demands.

This comprehensive (Taslimasa et al., 2023) survey examines the potential of the Internet of Vehicles (IoV) in enhancing driving experiences and safety, while also emphasising the vulnerabilities that can be manipulated by hostile adversaries. The text classifies security assaults, detailing their techniques, consequences, and targets. Additionally, it suggests a strategy for mitigating attacks called Intelligent Devices (ID), which relies on a detection-based approach. The work presents a quantitative risk assessment (QI) mechanism for evaluating and comparing Intrusion Detection Systems in Internet of Vehicles (IoV) environments. This survey (Ceviz et al., 2023) investigates the capacity of IoV to improve driving experiences and safety, while also highlighting the vulnerabilities that can be exploited by hostile attackers. The text categorizes security attacks, providing a comprehensive analysis of their methodologies, repercussions, and intended victims. In addition, it proposes a method for reducing threats known as Intelligent Devices, which depends on a detection-oriented approach. The study introduces a quantitative risk assessment (QI) mechanism to analyze and compare IDSs in IoV environments.

This critical review (Khraisat and Alazab, 2021) analyzes intrusion detection systems for the Internet of Things, focusing on methodology, deployment, validation strategies, datasets, and technology. It suggests the need for innovative IDS with minimal false alarms, flexibility for complex networks, and machine learning and deep learning methodologies. Future IDS should incorporate autonomous configuration, automatic optimization, inherent protection, and self-repair capabilities. This evaluation aids security researchers in creating IDS specifically for IoT communication. Federated Learning (FL) has been increasingly popular in the field of information and data security because of its decentralized, collaborative, and privacy-preserving characteristics. Nevertheless, there is a lack of a comprehensive and organized summary.

This systematic review (Hernandez-Ramos et al., 2023) seeks to address this deficiency by offering a thorough comprehension of FL-enabled IDS techniques, encompassing prominent Machine Learning (ML) models, datasets, aggregation functions, and implementation libraries (Yang et al., 2022). Additionally, it outlines obstacles and potential areas for future investigation, functioning as a point of reference for subsequent research endeavors.

These related works are compared to our main contribution based on the studied datasets, as illustrated in Table 2.

We aim to clarify the number of datasets utilized in previous research focused on detecting intrusions or anomalies in UAV networks and IDS. We examined the scientific studies detailed in Table 2, comparing the number of datasets referenced in earlier research with those

**Table 2**  
Comparative study of datasets for IDS.

| Datasets                      | No. of Datasets | MQTT-IoT-IDS2020 | MAVLINK dataset | UAVCAN | WSN-DS dataset | DJI Dataset | PX4 Dataset | ISCX 2012 | DEFCON-10 | UKM-IDS20 | Kyoto | ALFA | CAIDA | NSL-KDD | Ton-IoT | Bot-IoT | Edge-IIoTset | AWID3 | AWID2 | UAV Attacks | CIC-IDS2018 | CIC-IDS2017 | UNSW-NB15 | KDD Cup 99 | DARPA 99 |
|-------------------------------|-----------------|------------------|-----------------|--------|----------------|-------------|-------------|-----------|-----------|-----------|-------|------|-------|---------|---------|---------|--------------|-------|-------|-------------|-------------|-------------|-----------|------------|----------|
| Ajibuwa et al. (2023)         | 16              | ■                | □               | ■      | ■              | □           | □           | ■         | □         | □         | □     | □    | □     | ■       | ■       | ■       | ■            | ■     | ■     | ■           | ■           | ■           | ■         | ■          | ■        |
| Karopoulos et al. (2022)      | 5               | □                | □               | ■      | □              | □           | □           | □         | □         | □         | □     | □    | □     | □       | □       | □       | □            | ■     | ■     | ■           | □           | □           | □         | ■          | □        |
| Kheddar et al. (2023)         | 8               | □                | □               | □      | □              | □           | □           | □         | □         | □         | ■     | □    | □     | ■       | □       | ■       | ■            | ■     | □     | □           | ■           | ■           | ■         | ■          | □        |
| Momand et al. (2023)          | 11              | □                | □               | □      | □              | □           | □           | ■         | □         | □         | □     | □    | □     | ■       | ■       | □       | □            | □     | □     | □           | ■           | ■           | ■         | ■          | ■        |
| Taslimasa et al. (2023)       | 10              | □                | □               | ■      | □              | □           | □           | ■         | □         | □         | □     | □    | □     | □       | ■       | ■       | □            | ■     | ■     | □           | ■           | ■           | ■         | ■          | □        |
| Ceviz et al. (2023)           | 9               | □                | □               | □      | □              | □           | □           | □         | □         | □         | ■     | □    | □     | □       | ■       | ■       | □            | □     | □     | ■           | ■           | ■           | ■         | □          | □        |
| Khraisat and Alazab (2021)    | 8               | □                | □               | □      | □              | □           | □           | ■         | □         | □         | □     | □    | ■     | □       | □       | ■       | □            | ■     | ■     | □           | ■           | ■           | □         | ■          | ■        |
| Hernandez-Ramos et al. (2023) | 7               | ■                | □               | □      | □              | □           | □           | □         | □         | □         | □     | □    | □     | □       | ■       | ■       | □            | ■     | ■     | □           | ■           | ■           | □         | □          | □        |
| Yang et al. (2022)            | 9               | □                | □               | □      | □              | □           | □           | ■         | □         | □         | ■     | □    | □     | □       | ■       | ■       | □            | □     | □     | □           | ■           | ■           | ■         | ■          | ■        |
| Current Study                 | 24              | ■                | ■               | ■      | ■              | ■           | ■           | ■         | ■         | ■         | ■     | ■    | ■     | ■       | ■       | ■       | ■            | ■     | ■     | ■           | ■           | ■           | ■         | ■          | ■        |
| No. of Features               | 33              | 4C               | 4C              | 5C     | N/A            | N/A         | N/A         | 14        | N/A       | 41        | 24    | 5C   | 4C    | 46      | 80      | 46      | 61           | 46    | 96    | 4C          | 81          | 80          | 49        | 42         | 41       |
| No. of Attacks                | 6               | 2                | 3               | 3      | N/A            | N/A         | 4           | 4         | N/A       | 8         | 2     | 4    | N/A   | 22      | 12      | 5       | 14           | N/A   | N/A   | 3           | 5           | 15          | 19        | 4          | 4        |

\*C- means the main category for features.

used in the current investigation. The highest number of datasets in prior studies was 16, while the lowest was 5. In contrast, the current study utilized 24 datasets. Additionally, we compared the number of features in these datasets, which are crucial for intrusion detection and widely used in both academic research and real-world applications. Some studies emphasize reducing the number of features to enhance the effectiveness of intrusion detection. This study also included a comparison of the number of attacks present in each dataset.

### 3. UAVs communication systems and IDS architectures: Background

#### 3.1. UAVs communication systems

The field of unmanned aerial vehicle communications involves the transfer of data between a UAV and a specified endpoint, sometimes referred to as UAV-2-X communication (Xiaoning, 2020). Within this specific area, we present an introductory explanation of UAV-2-X communications. This request relates to the explanation of the communication architectures, networks created by unmanned aerial vehicles, and the routing protocols used within these networks. In the following discussion, we aim to provide a comprehensive explanation of the commonly acknowledged communication protocols (Gargalakos, 2021). Various forms of communication are employed in UAV-2-X systems. During a flying operation, an unmanned aerial vehicle initiates communication with multiple organizations. The exhibit (Figs. 2, 3, and 4) provided below depicts the categorization of UAV-2-X communications, which comprises four discrete endpoints.

##### 3.1.1. Mono-UAV communication systems

The establishment of effective communication channels between unmanned aerial vehicles and ground control stations plays a pivotal role in the successful execution of UAV missions. UAVs depend on this particular style of communication as their principal means of interaction (Wu et al., 2021). The Ground Control Station serves as a platform for the exchange of information between unmanned aerial vehicles through uplinks and downlinks, as shown in Fig. 3, thereby enabling the surveillance of aerial traffic and the execution of flying missions (Daud et al., 2022; Restas, 2023). This study investigates the types of datasets utilized in unmanned aerial vehicle communications. The term “control traffic” refers to the management and oversight of

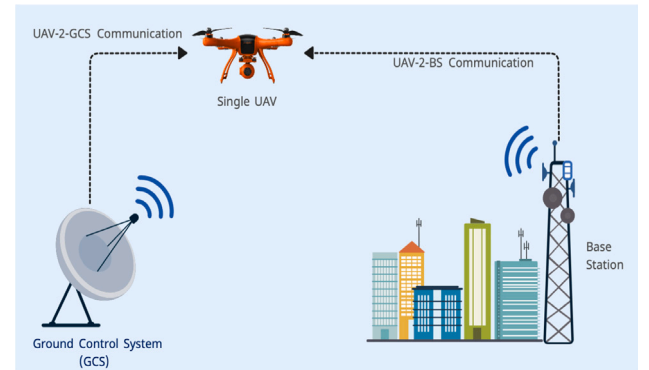


Fig. 3. Mono UAV communication.

directives related to control and monitoring. Specifically, this pertains to directives that are tailored to the job at hand and the up-to-the-minute status updates of UAVs, such as telemetry data and battery levels, among other relevant information (Al-Wathinani et al., 2023). The coordination of traffic refers to the management of collaboration among several Unmanned Aerial Vehicles during flight missions and the execution of duties that are independent of the Ground Control Station, such as collision avoidance procedures (Mishra et al., 2022). The data collected by the onboard sensors is encapsulated within the sensing traffic and subsequently communicated to GCS. It is noted that the many forms of communication between UAV-2-GCS use wireless technologies, as in Fig. 4. Additionally, these wireless technologies are often lacking in security measures, rendering them susceptible to both passive and active attacks.

##### 3.1.2. Swarm-UAVs communication systems.

The communication between unmanned aerial vehicles, also known as air-to-air communication, occurs during flying operations that involve the use of numerous UAVs (Hentati and Fourati, 2019). In such situations, UAVs engage in collaborative and coordinated activities using wireless technologies that have low power consumption, such as Bluetooth and Zigbee. These technologies enable the UAVs to share information either directly or through multi-hop wireless connections. The utilization of UAV swarms offers several notable benefits, such

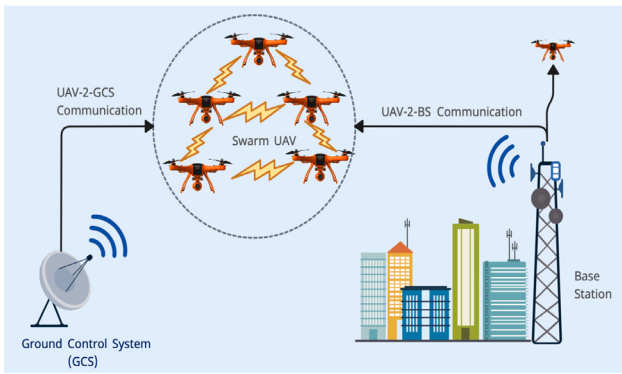


Fig. 4. Swarm UAV communication.

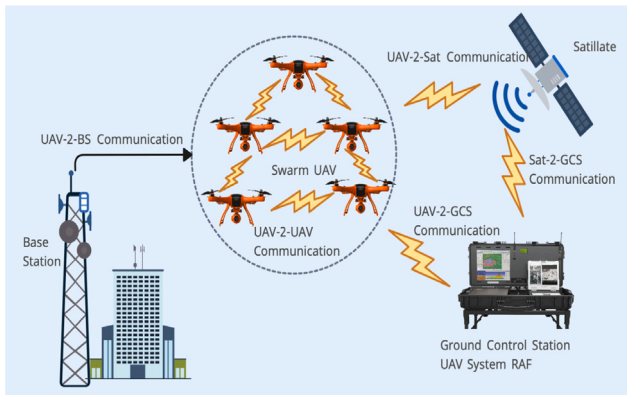


Fig. 5. Network UAV communication.

as the reduction of task completion time, expansion of coverage area, and decreased operational costs. In this scenario, a solitary UAV functions inside a network of UAVs to exchange data and successfully execute the intended aerial mission. Nevertheless, the connection between UAVs suffers from a very limited throughput and transmission bandwidth (Hentati and Fourati, 2021).

### 3.1.3. Networked-UAVs communication systems

**Unmanned Aerial Vehicle to Satellite Communication:** During Beyond Line-of-Sight (BLOS) missions, it is imperative for the operator to accurately determine the position of the UAV in order to ensure safe navigation (Tsao et al., 2022). Consequently, UAVs have the capability to establish a satellite-based communication link in order to acquire and communicate their current GPS coordinates to the GCS in real-time. In addition, satellite communications offer major advantages in scenarios when there is a lack of established infrastructure over considerable distances, facilitating dependable communication with substantial transmission bandwidth. Furthermore, the utilization of commercial satellite communications can be employed for the purpose of managing unmanned aerial vehicles (see figure as in Fig. 5). Nevertheless, these devices are characterized by their substantial energy consumption and the considerable expenses associated with their upkeep. Additionally, their utilization may lead to the emergence of significant latency concerns.

### 3.2. Intra vs inter UAVs communication

Intra-UAV communication refers to the communication between the different interlinked embedded modules within any UAV. Indeed, seamless communication between the embedded modules is fundamental to ensure cohesive and efficient UAV operation (Shrestha et al.,

2021; Tsao et al., 2022). The intra-UAV communication architecture comprises several integrated modules, each serving a specific function and contributing to the overall functionality of the aerial vehicle. Buscan protocol can be considered as a potential communication protocol that could handle the communication between the diverse embedded modules within the UAV system. Among the UAV-embedded modules, we highlight the following:

- **Navigation Module:** The navigation module serves as the core component responsible for flight path planning and real-time navigation. Through a dedicated communication protocol (mainly buscan), it interfaces with other modules, providing continuous updates on positional data, route optimization, and environmental feedback.
- **Sensing and Perception:** Embedded sensors, including cameras, LiDAR, and other environmental perception devices, communicate intricately with the navigation module. They relay crucial information about the UAV's surroundings, including terrain mapping, obstacle detection, and environmental changes. This data exchange is pivotal for real-time decision-making and adaptive flight adjustments.
- **Payload and Task-Specific Modules:** Modules dedicated to specialized tasks or payloads, such as imaging, payload delivery, or specific scientific instrumentation, communicate with the navigation and sensing modules. This interaction ensures coordinated actions, aligning the UAV's movement with the task objectives and leveraging real-time data to optimize task performance.
- **Communication and Control:** A dedicated communication and control module facilitates inter-module communication, acting as a nexus for data exchange. This module manages data flow, orchestrating seamless communication between different modules while ensuring a coherent and synchronized operation of the UAV system.

Effective communication between unmanned aerial vehicles (UAVs), either within groups or between individual UAVs, is necessary for promoting cooperative behavior, improving situational awareness, making the best use of resources, and completing mission objectives in areas like search and rescue, surveillance, environmental monitoring, and autonomous aerial missions.

Inter-UAV communication refers to the exchange of data and information between multiple UAVs. This communication plays a pivotal role in various applications, including collaborative tasks, coordinated missions, collision avoidance, and distributed sensing. Inter-UAV communication enables UAVs to share information such as location, sensor data, mission objectives, and environmental observations, facilitating coordinated actions and enhancing the overall capabilities of the UAV network. Techniques such as ad-hoc networking, and swarm intelligence are essential in establishing robust and reliable communication links between UAVs in dynamic and challenging environments.

Within inter-UAV communication, there are two primary types (see as in Fig. 6:

- **Inter-UAV Group Communication:** UAVs are organized into groups, and each group functions as a FANET. Inter-UAV Group Communication involves communication among groups or clusters of UAVs. It focuses on the interaction and collaboration between distinct sets of UAVs working together to accomplish shared objectives. Group communication allows for the coordination of tasks, distribution of responsibilities, and synchronization of actions among different subsets of UAVs within a larger network.
- **Intra-UAV Communication:** In contrast, intra-UAV communication pertains to the communication that occurs within an individual UAV. This involves the internal exchange of data and commands among the various components and systems within a single UAV platform. Intra-UAV communication ensures smooth operation, coordination between onboard sensors and systems, and effective execution of tasks by the UAV itself. as in Fig. 7:

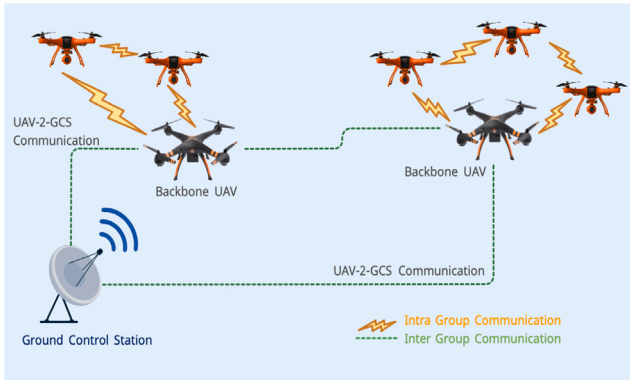


Fig. 6. Intra and Inter UAVs communication.

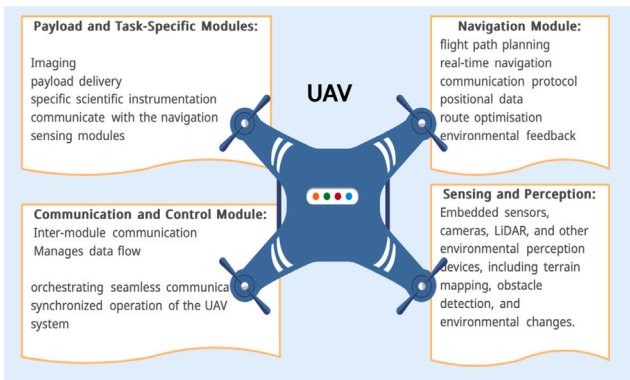


Fig. 7. Inter UAVs communication.

Fig. 7 presents the different embedded modules within UAV as well as the Intra and Inter UAV communication (see Table 3).

**Intra-UAV Attacks:** encompass malevolent actions occurring within a singular (UAV), typically directed toward compromising the integrity or functionality of its internal software or hardware components. The aforementioned assaults have the potential to undermine the integrity, availability, or secrecy of the UAV. Instances of intra-unmanned aerial vehicle assaults encompass as in :

**Inter-UAV Attacks:** refer to instances where one (UAV) gains illegal access to or manipulates another UAV, either through the actions of another UAV or an external party. The aforementioned attacks have the capability to specifically target the communication links, control systems, or sensor data of (UAVs). Instances of inter-unmanned aerial vehicle attacks encompass as in :

### 3.3. IDS basic concepts

An Intrusion Detection System (IDS) is a security instrument that has been specifically developed to observe and scrutinize network or system activity in order to identify any potentially suspicious or harmful behavior. The primary objective of this system is to identify and promptly address any security vulnerabilities, hence safeguarding enterprises from cyber threats. There exist two primary categories of IDS, namely Network-based Intrusion Detection Systems (NIDS) and Host-based Intrusion Detection Systems (HIDS) (Mohammed et al., 2023). Besides that, detection techniques could be based on protocol analysis, behavioral analysis, signature-based IDS, and anomaly-based IDS. The categories of IDS are clearly illustrated in Fig. 8 A.

- **Network-Based IDS (NIDS):** Monitors network traffic in real-time to detect suspicious patterns or signatures, typically positioned at strategic points within the network.

Table 3

Intra-UAV attacks (Ajibuwa et al., 2023; Karopoulos et al., 2022).

| No. | Type of Attack                  | Description                                                                                                                                                                                                                                                                                                                                                                    |
|-----|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Malware infection               | The malware infection attack pertains to the deliberate infiltration of the unmanned aerial vehicle's operating system with malicious software, with the intention of impeding its normal functioning or illicitly acquiring confidential information.                                                                                                                         |
| 2.  | Logic bombs                     | The utilization of logic bombs encompasses the insertion of malevolent programming that initiates detrimental actions contingent upon particular conditions. As an illustration, a logic bomb has the potential to be programmed in such a way that it triggers the detonation of the unmanned aerial vehicle's explosives upon its traversal of a specific geographic region. |
| 3.  | Denial-of-service (DoS) attacks | The denial-of-service (DoS) assaults refer to a malicious act of inundating the resources of (UAV) to the point where it becomes unresponsive to valid requests. One potential approach to achieve this objective involves inundating (UAV) with an excessive amount of network traffic or transmitting a substantial volume of requests to the UAV (see Table 4).             |

Table 4

Inter-UAV attacks (Kheddar et al., 2023; Momand et al., 2023).

| No. | Type of Attack                     | Description                                                                                                                                                                                                                                                                                                                                                           |
|-----|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | GPS spoofing                       | refers to the act of changing the Global Positioning System signals of an unmanned aerial vehicle to modify its position or trajectory. This capability has the potential to induce a UAV malfunction, resulting in a crash or deviation from its intended flight path.                                                                                               |
| 2.  | Command and control (C2) hijacking | The act of command and control (C2) hijacking refers to the unauthorized acquisition of control of the command and control system of an unmanned aerial vehicle, hence enabling the issuance of commands that are not allowed. This capability enables the manipulation of the unmanned aerial vehicle to initiate takeoff, execute landing, or alter its trajectory. |
| 3.  | Data tampering                     | The act of data tampering encompasses the interception and modification of sensor data to deceive the control systems of UAVs. This technique has the potential to induce a false perception of location or alter the perceived situation of the UAV, hence creating a discrepancy between the actual and perceived states.                                           |

- **Host-Based IDS (HIDS):** Operates on individual devices or hosts, analyzing activities such as file modifications, logins, and process executions to identify potential intrusions (Momand et al., 2023).
- **Behavioral Analysis:** Involves studying behavioral patterns to identify deviations. Machine learning and AI are increasingly used for behavioral analysis to detect unknown or evolving threats.
- **Protocol Analysis:** Inspects network packets to detect malicious activities based on analyzing the protocols being used.



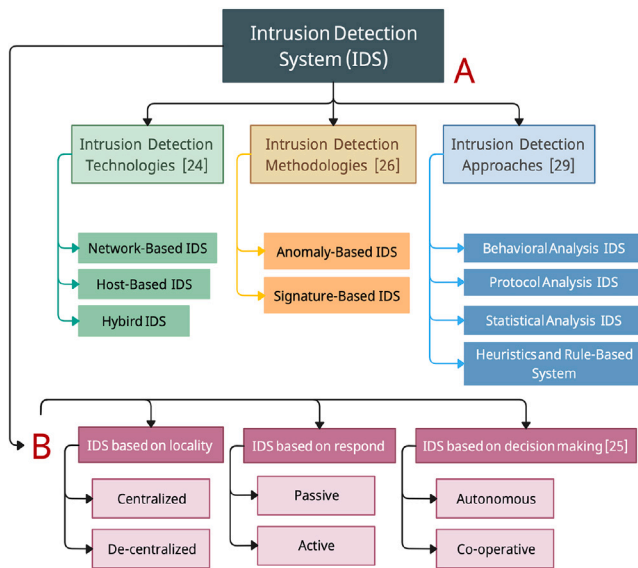


Fig. 8. Categories of IDS A and B.

- **Heuristics and Rule-Based Systems:** Defining rules or heuristics to identify suspicious behavior, allowing for customized detection of specific threats (Khraisat and Alazab, 2021).
- **Statistical Analysis:** Analyzing statistical data to identify anomalies in network traffic or system behavior.
- **Signature-Based IDS:** These systems use predefined patterns or signatures to detect known threats (Karopoulos et al., 2022). They compare incoming traffic or system behavior against a database of signatures to identify known attack patterns.
- **Anomaly-Based IDS:** Anomaly detection involves establishing a baseline of “normal” behavior and flagging any deviations from this baseline (Ajibuwa et al., 2023). These systems detect previously unseen threats by identifying abnormal activities.
- **Hybrid IDS:** Combining both signature-based and anomaly-based approaches, hybrid IDS aims to leverage the strengths of both methods to improve accuracy and reduce false positives.

To create a more persuasive classification, it would be beneficial to adopt a more nuanced and flexible framework that accounts for the multi-faceted nature of IDS that involves a combination of dimensions in Fig. 8 B, such as (Yang et al., 2022): Based on locality: Centralized IDS, De-centralized IDS. Response Strategy: Passive detection, active response. Based on decision making: Autonomous IDS, Co-operative IDS (Kheddar et al., 2023). Such a multi-dimensional classification approach would better capture the diversity and complexity of modern IDS, providing clearer distinctions and a more comprehensive understanding of the different systems. After that, we provided a new taxonomy that took into consideration these new dimensions.

### 3.4. Integration of IDS within UAVs systems

In recent years, Network Intrusion Detection Systems have demonstrated increased use and efficacy. Overviews UAVs communication systems, IDS architectures, and integration within UAVs communication systems. Artificial Intelligence approaches are commonly employed in IDS within the domain of cybersecurity (Nuaimi et al., 2023). Within an autonomous system, a manually developed test is considered a fundamental method of testing. Artificial Intelligence (AI) is involved in the examination and analysis of scenarios generated by a human tester. Consequently, the effectiveness of AI is assessed. Hence, there have been proposals for datasets that may be used to compare and evaluate various Network Intrusion Detection Systems. A (NIDS) comprises three

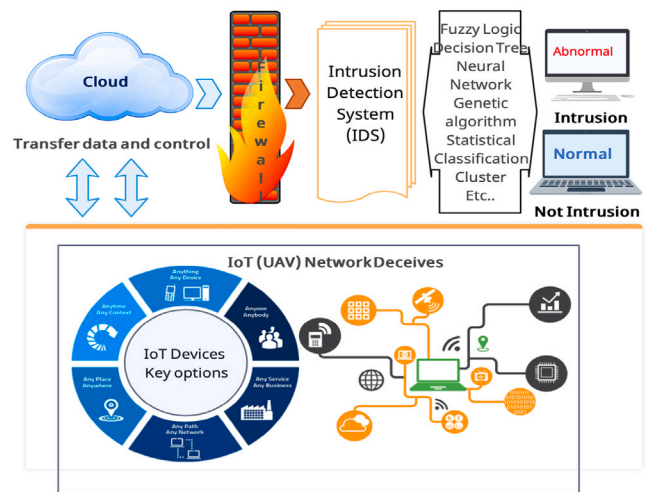


Fig. 9. IDS AI integration within UAVs.

fundamental components, namely an agent, an analysis engine, and a response module. The primary role of the agent is to gather data from the network through the process of event monitoring. On the other hand, the analysis engine and the reaction module have the respective responsibilities of detecting indications of intrusion, producing alerts, and processing the outcomes obtained from the analysis engine (see Fig. 9). However, it is important to acknowledge that adversaries have concurrently devised diverse attack methodologies to circumvent these detection technologies. Furthermore, it should be noted that conventional NIDS may not be well-suited for the intricate network layers found on the Internet of Things. The involvement of Unmanned Aerial Vehicles introduces heightened complexity, as drone operations exhibit greater intricacy and necessitate enhanced security measures compared to other networks, owing to their important uses. Because of the above development, datasets for UAVs were created to make it easier to keep an eye on their activities in line with the tasks they are supposed to do and the threats they face.

## 4. Datasets toward IDS for UAVs communication systems: Characteristics and taxonomy

### 4.1. Proposed taxonomies of datasets toward IDS

In this section, we suggest taxonomy IDS datasets according to their specific domain focus, datasets customized for specific contexts such as UAVs and IoT, and datasets suitable for intra and inter-UAV interactions.

#### 4.1.1. Proposed taxonomy of available IDS datasets

As the first taxonomy, we propose to categorize IDS datasets based on their specific domain focus, enabling researchers and practitioners to identify datasets tailored to particular environments (UAVs, IoT) or select more general-purpose datasets for broader evaluation and analysis of IDS techniques as shown in Fig. 10.

- **IDS Datasets for UAVs:** This category includes datasets specifically curated for Intrusion Detection Systems deployed in Unmanned Aerial Vehicles or drones. These datasets are tailored to assess the security and vulnerabilities of UAV networks, considering the specific challenges and constraints faced by airborne systems. Besides that, these datasets capture network traffic, system logs, or communication patterns unique to UAV-based systems. The datasets under this category focus on capturing the intricacies of communication and network activities within UAV networks.

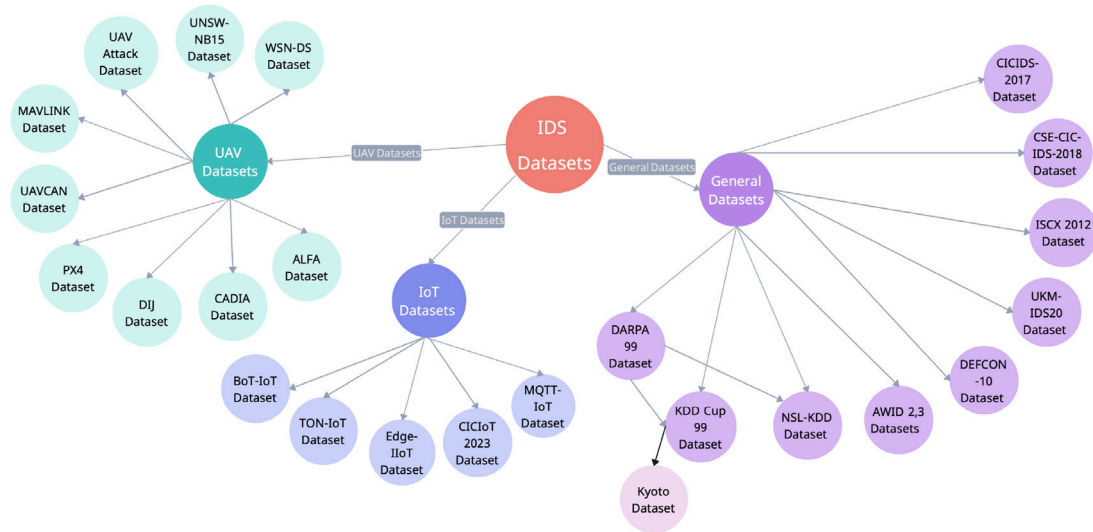


Fig. 10. Taxonomy of IDS datasets.

They might encompass various UAV-specific protocols such as MAVLINK protocol, flight control data, sensor information, and communication between ground stations and UAVs.

- **IDS Datasets for IoT:** This category comprises datasets intended for IDS deployed in IoT environments. These datasets capture network traffic, device interactions, and communication protocols within IoT ecosystems. IoT-focused datasets encompass a diverse range of IoT devices, sensors, and communication protocols prevalent in smart homes, industrial IoT, healthcare, or smart city settings. They may include data from sensors, actuators, gateways, and IoT devices communicating over various protocols like MQTT, CoAP, or Zigbee.
- **General Use IDS Datasets:** This category encompasses datasets that serve as foundational resources for IDS systems across diverse domains beyond UAVs or IoT. These datasets are versatile and applicable to a wide array of network security scenarios. General-use datasets consist of network traffic, system logs, or simulated attack scenarios sourced from various environments like enterprise networks, academic settings, or simulated network traffic. They are diverse in attack types, network protocols, and system behaviors. These datasets are instrumental in evaluating and benchmarking IDS models across general network security scenarios, providing a baseline for comparison and analysis.

#### 4.1.2. Proposed specific taxonomy of IDS datasets suitable for intra and inter-UAV communications

In recognition of the intricate communication structures within UAV networks, we introduce a specialized taxonomy aimed at categorizing IDS datasets to better capture the nuances of Intra-UAV and Inter-UAV communications. The Taxonomy of IDS Datasets for Intra and Inter-UAVs is clearly illustrated in Fig. 11. Accordingly, the proposed taxonomy considers datasets that can be used for IDS that target UAVs intra and inter-communications.

#### 4.1.3. Proposed specific taxonomy of IDS datasets within UAV systems

In addition to the aforementioned taxonomies, we introduced a supplementary classification system specifically targeted at datasets relevant to UAVs. This secondary taxonomy aims to provide a more specialized breakdown of IDS datasets catering to the unique communication structures within UAV networks. Within this taxonomy, we delineated distinct categories to encapsulate the intricacies of UAV communications, categorizing them into two fundamental types: Intra-UAV communication and Inter-UAV communication.

The subdivision of IDS datasets for UAVs includes a nuanced classification system that addresses the diverse communication aspects within UAV networks. The first subcategory, 'IDS Datasets for Intra-UAV Communication,' is designed to capture and analyze the interactions among the internal components or modules within a single UAV system. This classification encapsulates datasets focusing on communication protocols, sensor data exchanges, and internal network traffic within a singular UAV entity.

Further refining this taxonomy, the 'IDS Datasets for Inter-UAV Communication' category encompasses two distinct subcategories to comprehensively represent the complexities of communication networks involving multiple UAVs. These subdivisions cater to the specific communication paradigms present in UAV networks. The first subcategory, 'UAV-GCS Communications,' concentrates on datasets that portray the communication dynamics between UAVs and Ground Control Stations. These datasets encompass communication protocols, telemetry data, command and control interactions, and information exchange between UAVs and their designated ground control entities as illustrated in Fig. 12.

The second subcategory, 'UAVs-UAVs Communications,' delves into datasets representing communication scenarios among multiple UAVs operating within a network or collaborative mission framework. This classification encapsulates datasets that portray interactions, data sharing, coordination protocols, and collaborative tasks among interconnected UAVs.

By refining our taxonomy to include this comprehensive breakdown specific to UAV-based datasets, we aim to offer a detailed framework that aptly represents the varied communication landscapes within UAV networks, thereby facilitating more targeted and specialized evaluations of IDS in UAV-centric scenarios.

#### 4.2. IDS datasets specific for UAVs communication systems

We have categorized IDS datasets specifically tailored for UAV communication systems into two distinct groups. The first category centers on datasets dedicated to assessing intrusions within Intra-UAV communication systems, exemplified by the UAVCAN, UAV-Attack PX4, and DJI IDS datasets. The second category is specifically geared toward identifying intrusions within Inter-UAV communication systems, employing various datasets such as MAVLINK, ALFA, UNSW-NB, WSN-DS, and CADIA".

In the following subsections, we overview the UAV datasets dedicated to IDS within Intra-UAVs communication followed by the datasets useful for IDS within Inter-UAVs communication.

##### First: Intra-UAVs communication datasets

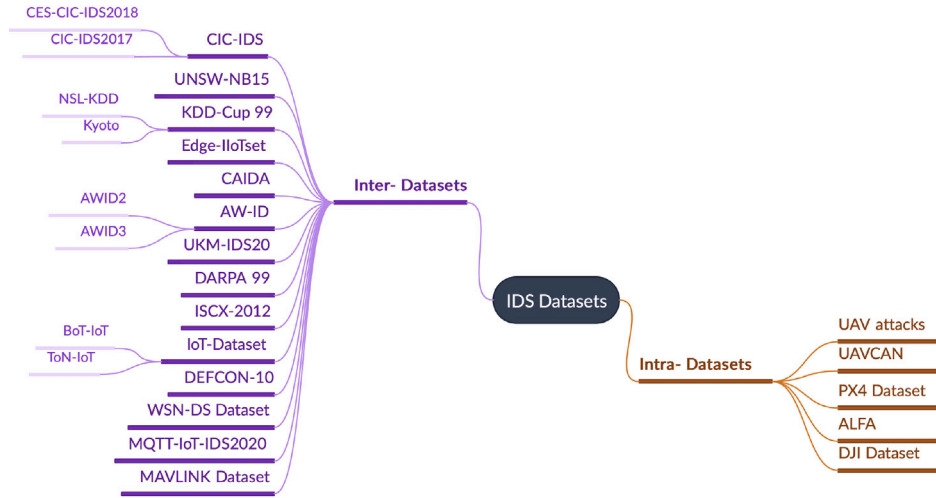


Fig. 11. Intra and Inter IDS datasets taxonomy.

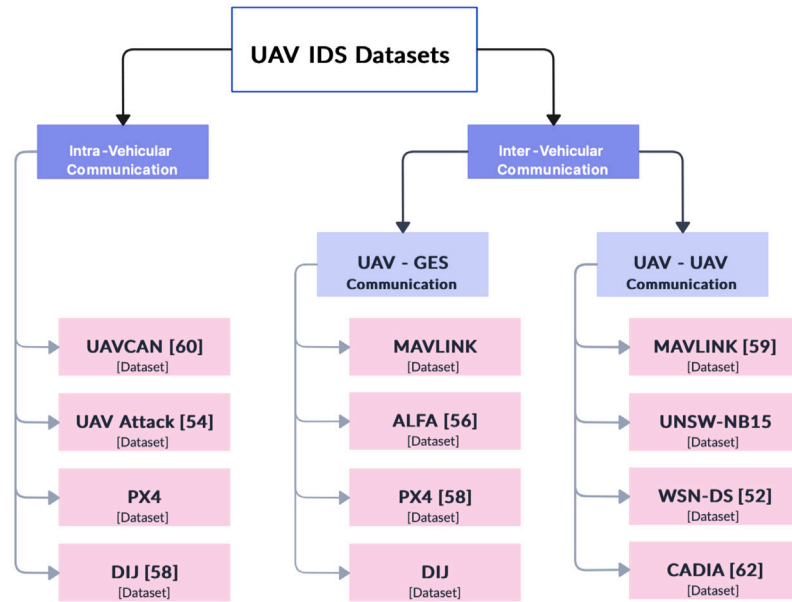


Fig. 12. IDS datasets taxonomy for UAVs.

#### 4.2.1. UAVCAN IDS dataset

The UAVCAN Dataset comprises a compilation of log files encompassing data obtained from UAVCAN nodes (Ravi and El-Sharkawy, 2019). It focuses on capturing and analyzing network traffic within UAVCAN, a lightweight protocol designed for reliable intra-UAV communication. UAVCAN operates on an open-source basis. The dataset encompasses data collected from a diverse range of UAVs and sensors. It comprises various network packet captures, including message exchanges, command flows, and communication patterns within UAVCAN networks. It can be effectively employed for the purpose of training machine learning models rule-based approaches, to enable the development and validation of algorithms to detect anomalies or deviations from expected behavior within UAVCAN networks. Besides that, it allows the creation of signatures or patterns of known attacks or malicious activities, aiding in the identification and prevention of similar threats in real-time.

#### 4.2.2. UAV attack IDS dataset

The UAV Attack Dataset: is a limited collection of datasets that includes ad-hoc network traffic (Tlili et al., 2023; Ajakwe et al., 2024).

The focus of this study is on environments pertaining to Unmanned Aerial Vehicles (UAVs), and the development of the research was facilitated through the utilization of five distinct simulation platforms. The dataset encompasses two distinct attack scenarios: Global Positioning System spoofing and jamming. Despite being the sole option accessible for researching UAV IDS.

#### 4.2.3. PX4 IDS dataset

The PX4 dataset, often known as Dataset 21, is the subject of discussion. The dataset was collected using a cost-effective flight controller board known as Pixhawk, which is suitable for both research and industry applications (Shrestha et al., 2021). The board incorporates a combination of IMU sensors, a magnetometer, flight control components, monitoring components, an ARM processor, and a micro SD card to capture the necessary data from the IMU sensors and flight information. The individuals who created the initial Pixhawk flight controller have started working on developing the open-source autopilot software known as PX4 autopilot. Researchers can utilize this material as a reliable reference for the sake of data-driven methodologies. The database consists of flight logs with real flight data that users from all

over the world are gathering and uploading. The PX4 team is now in charge of maintaining and managing the logs. The database contains a substantial number of flight logs pertaining to a wide range of vehicles and diverse iterations of the Pixhawk platform.

#### 4.2.4. DJI IDS dataset

The DJI Dataset also referred to as Dataset 22, is a collection of data that DJI, a well-known company in the unmanned aerial systems (Ghurab et al., 2021). In order to generate the dataset, we simulated the DJI-Phantom and DJI-Mavic models, and afterwards obtained a freely available 3D model of the DJI-Inspire. Subsequently, a script was developed with the purpose of generating randomized photographs of the drone. The process involves the utilization of a three-dimensional (3D) model of a drone, to which textures and High Dynamic Range Images (HDRI) are applied, resulting in the generation of rendered photographs. The provided visual representations depict computer-generated images produced through the utilization of a simulation developed within the Blender software (Fadhila et al., 2023). The many varieties of drones, namely the DJI Phantom, DJI Mavic, and DJI Inspire, are depicted. Finally, the class labeled as “no drone” is presented, devoid of any depictions of drones, and exclusively featuring miscellaneous backdrop visuals. The number 23 is the subject of discussion.

### Second: Inter-UAVs communication datasets

#### 4.2.5. MAVLINK IDS dataset

The MAVLINK dataset, consisting of flying system parameters, was obtained by utilizing the PX4 autopilot and Gazebo robotics simulator (Ravi and El-Sharkawy, 2019). The PX4 autopilot firmware is widely utilized in numerous unmanned aerial vehicles as an open-source solution. Hence, the MAVLINK dataset encompasses comprehensive information, specifically pertaining to a vast array of unmanned aerial vehicles (UAVs). The dataset has two distinct groupings of data samples. The initial category pertains to the parameters associated with a routine flight, assuming the absence of any hostile actions (Jullian et al., 2022). The second category pertains to the parameters of a flight that is subjected to a spoofing assault, which is also referred to as a spoofed flight. It is crucial to recognize that the spoofing assault lasts for 30 s, whereas the flight time varies between 10 and 30 min. This may indicate a lack of equilibrium in the dataset. Next, the author recommends focusing only on the flight metrics that exhibit significant variability. These parameters may be separated into three distinct groups: GPS coordinates, position and orientation records, and system and control status.

#### 4.2.6. CADIA IDS dataset

The CAIDA dataset comprises two distinct datasets encompassing both normal and assault traffic. Consequently, we can achieve near-perfect detection scores for the aforementioned datasets (Fadhila et al., 2023). The AMP DDoS detection method exhibits a higher CID value due to its reduced occurrence of false-positive outcomes. The utilization of a misuse dictionary yields optimal detection outcomes, while the application of an anomaly dictionary also contributes to effective detection. The detection methods based on Wavelet and MPMP utilize one-dimensional vectors representing traffic attributes. Consequently, a total of 16 distinct outcomes are derived through the utilization of these methodologies.

#### 4.2.7. WSN-DS dataset

The dataset is referred to as WSN-DS. The dataset employed for the identification of network assaults on UAV swarms is referred to as WSN-DS (Ghurab et al., 2021). The system operates by employing four distinct forms of (DoS) attacks, including blackhole, grayhole, scheduling, and flooding, alongside its regular operational behavior. The Low Energy Aware Cluster Hierarchy (LEACH) routing protocol takes into account the peculiarities of Wireless Sensor Networks (WSN)

throughout its application. The scheduling assault was excluded from consideration in this work due to its infrequent occurrence in UAV swarms, unlike other types of attacks. Therefore, network behavior can be categorized as either normal or subject to blackhole, grayhole, and flooding attacks. Additionally, the data undergoes min-max normalization.

#### 4.2.8. UNSW-NB 15 dataset

The UNSW-NB 15 dataset: is a network traffic dataset created in 2015, focuses on current minimal footprint invasions. There is a total of 2.5 million records with 49 features. Distinct traits that can be categorized into multiple groups (Ahmad et al., 2022; Khraisat and Alazab, 2021). It was generated in the context of a real-world environment, making it potentially suitable for IDS within UAV systems. Furthermore, the UNSW-NB15 contains network traffic data captured in a controlled university lab environment. It consists of normal traffic as well as various types of attacks, including DoS, probing, infiltration, and more. The dataset comprises. These features include basic information about network connections such as source and destination IPs, service and protocol types, packet sizes, and more. This variety of features can potentially provide insights into various network activities within UAV systems. For IDS within UAV systems, the UNSW-NB15 dataset can be used to train and test machine learning models for intrusion detection (Escorcia-Gutierrez et al., 2023). By leveraging the dataset's features and attack labels, algorithms can be developed to detect and classify suspicious network activities that may pose security risks to UAV operations. UAV systems often operate in dynamic and challenging environments, which may have different network configurations and constraints compared to traditional networks. Thus, the dataset's adaptability to such environments needs evaluation. While the UNSW-NB15 dataset provides a sizable collection of network traffic data with various attack scenarios, its direct applicability to UAV systems requires further investigation and potential enhancements to better represent the unique characteristics and security challenges of the UAV communication network.

#### 4.2.9. ALFA IDS dataset

Dataset ALFA provides information on various defect kinds in the control surfaces of a fixed-wing (UAV) (Fadhila et al., 2023). The dataset is intended for research purposes related to defect Detection and Isolation (FDI) as well as Anomaly Detection (AD). At present, the dataset comprises processed data from a total of 47 autonomous flights. Among these flights, there are 23 instances of sudden full engine failure scenarios and 24 instances of sudden control surface (actuator) faults, encompassing seven different types. Furthermore, the dataset encompasses a substantial amount of raw data derived from flights that were conducted under three different modes: fully autonomous, autopilot-assisted, and manual.

## 5. IDS datasets

Based on our investigation, various datasets that serve as foundational resources for IDS are now available, each offering unique insights and challenges in enhancing system robustness. In the following, we present the results of our investigation.

### 5.1. Characteristics and main features of datasets toward IDS

Datasets used for IDS are essential for training, testing, and evaluating the effectiveness of intrusion detection algorithms and models. provides a summary of the most significant characteristics of IDS datasets. Next, provides a summary of the Main Features of IDS datasets. Here are the main characteristics and features of datasets:



- **Variety of Attacks:** A comprehensive IDS dataset should encompass various types of cyberattacks, including but not limited to Denial of Service (DoS), Distributed DoS (DDoS), probing attacks, malware, buffer overflow, SQL injection, and other common attack vectors (Neto et al., 2023). The diversity of supported attacks allows for a more thorough assessment of the IDS's capability to detect different types of threats.
- **Normal and Anomalous Data:** A robust dataset should contain both normal and anomalous network traffic or system behavior. Normal data represents legitimate user activities, while anomalous data includes instances of attacks or suspicious behavior. This variety helps in training the IDS to distinguish between normal and malicious activities accurately.
- **Realism and Volume:** The dataset should ideally mimic real-world network traffic or system logs, containing a substantial volume of data to reflect the complexity and scale of modern networks. Realistic datasets enable the IDS to learn from and adapt to genuine network patterns and cyber threats.
- **Labeling and Ground Truth:** Accurate labeling of data points with ground truth information about whether they represent normal or malicious activities is crucial. This labeled data is essential for supervised learning techniques, allowing IDS models to learn from known patterns and behaviors.
- **Temporal and Spatial Features:** Datasets may include temporal features, such as time stamps or sequential information, aiding in understanding the timing and sequence of events. Spatial features, like IP addresses, ports, or packet content, provide information about the spatial characteristics of network traffic.
- **Scalability and Diversity:** Scalable datasets that cover diverse network environments, such as different network topologies, protocols, and traffic distributions, help in testing IDS under varied conditions and ensure its adaptability across different network configurations.
- **Anonymization and Privacy Preservation:** To comply with privacy regulations and ethical considerations, datasets might be anonymized or sanitized to remove sensitive information while retaining the essential characteristics necessary for IDS training and evaluation (see Tables 5 and 6).

**Table 5**

IDS datasets characteristics (Taslimasa et al., 2023; Ceviz et al., 2023).

| No. | Characteristics                     | Description                                                                                                                                                                                   |
|-----|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Data Diversity                      | The presence of diversity inside the model contributes to its ability to effectively generalize across various network architectures and attack situations.                                   |
| 2.  | Realism and Authenticity            | The utilization of realistic data in the training and testing of the IDS model ensures that the model is exposed to and evaluated.                                                            |
| 3.  | Labeling and Ground Truth           | The establishment of a definitive and unambiguous reference point is of utmost importance in the process of training and assessing the precision of (IDS) models.                             |
| 4.  | Anomaly and Signature Data          | The utilization of signature data, which encompasses recognized attack patterns, holds significant value in the context of signature-based detection.                                         |
| 5.  | Temporal Dynamics                   | This approach facilitates the representation of the evolving characteristics of both typical and harmful actions during a given timeframe.                                                    |
| 6.  | Imbalanced Class Distribution       | A dataset exhibiting an imbalanced class distribution is representative of the real-world scenario and aids the model in effectively addressing the disproportionate distribution of classes. |
| 7.  | Scalability                         | This enables the evaluation of the performance of the IDS model in various deployment circumstances.                                                                                          |
| 8.  | Real-World Traffic Representation   | This encompasses routine operations, with a range of offensive actions and unauthorized infiltrations that are probable in real-world network settings.                                       |
| 9.  | Representative Network Environments | The presence of diversity within IDS models enables them to effectively adapt to a wide range of real-world scenarios.                                                                        |
| 10. | Dynamic Updates                     | IDS databases should be updated to reflect new attacks and variations in existing attack patterns as the threat landscape changes.                                                            |

## 5.2. General use IDS datasets

This section overviews several widely utilized datasets in the realm of IDS comprising DARPA 99+, KDD-Cup '99, Kyoto, NSL-KDD, DEFCON-19, UKM-IDS20, CSE-CIC-IDS2018, CICIDS-2017, ISCX 2012, and AWID 2.3, each offering diverse sets of network traffic, cybersecurity attacks, anomalies, or mixed data to facilitate the evaluation and development of IDS methodologies, algorithms, and models within varied (simulated and real-world) environments.

### 5.2.1. DARPA dataset

**DARPA:** The initial attempt to generate a dataset for Intrusion Detection Systems was undertaken by DARPA (Defense Advanced Research Projects Agency) in 1998 (Yang et al., 2022; Hnamte et al., 2023). This endeavor resulted in the creation of the KDD98 dataset, which falls within the domain of Knowledge Discovery and Data Mining (KDD). In the year 1998, the (DARPA) initiated a program in collaboration with the Massachusetts Institute of Technology (MIT) Lincoln Laboratory. The objective of this effort was to establish an extensive and authentic benchmarking environment for IDS, (Lincoln Laboratory, 1999). The dataset in question has been frequently challenged for its correctness and its capacity to account for real-life settings, despite its significant contribution to research on IDS. The datasets were obtained by employing a network of interconnected computers connected to the Internet to simulate a limited-access US Air Force post with a restricted personnel population.

### 5.2.2. Kddcup99 dataset

The KDDCup99 dataset: is widely recognized as the most often employed dataset in the field of Intrusion Detection Systems. A total of 4,900,000 assault records are accessible. The dataset utilized in this study was derived from the tcpdump dataset through a processing procedure. The aforementioned technology was initially introduced to the public during the 1998 DARPA ID challenge (Ajibuwa et al., 2023; Taslimasa et al., 2023; Ceviz et al., 2023). Each entry in the dataset is associated with 41 distinct features. Besides that, it includes different classes of data, such as normal traffic and various attack types like DoS (Denial of Service), Probe, R2L (Unauthorized access from a remote machine), U2R (Privilege escalation attempts), etc. While it is a comprehensive dataset, it is worth noting that the KDDCup 99 dataset has some limitations. One of the most discussed issues is its representation of attacks, which might not accurately reflect real-world scenarios due to its synthetic nature. In addition, due to the size and complexity of the dataset, preprocessing and feature selection are often necessary to handle its dimensions efficiently and reduce noise for effective IDS model building. Despite its limitations, the KDD Cup 99 dataset remains a benchmark dataset in the field of intrusion detection. However, due to its synthetic nature and evolving attack techniques, researchers often supplement it with other datasets or create more realistic datasets to address its limitations.

### 5.2.3. NSL-KDD dataset

The NSL-KDD dataset is an enhanced iteration of the dataset employed in the KDDCup99 competition (Ajibuwa et al., 2023). The

**Table 6**  
IDS datasets main features (Khraisat and Alazab, 2021; Yang et al., 2022).

| No. | Main Features               | Description                                                                                                                                                                                                                                              |
|-----|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Labeled Instances           | The process of labeling is of utmost importance in the context of supervised learning and serves as a means to assess the effectiveness of intrusion detection models. Table 3 provides a summary of the most significant Main Features of IDS datasets. |
| 2.  | Variety of Attack Scenarios | encompassing prevalent forms of attacks such as denial-of-service (DoS), distributed denial-of-service (DDoS), malware, and infiltration attempts employing various methodologies.                                                                       |
| 3.  | Temporal Aspects            | Temporal attributes, including the timing and order of occurrences, have significance in comprehending the dynamics of network activities.                                                                                                               |
| 4.  | Packet-Level Information    | This enables the model to effectively examine the content and arrangement of network packets in order to enhance the precision of intrusion detection.                                                                                                   |
| 5.  | Privacy Considerations      | Datasets can undergo a process of anonymization or misidentification in order to safeguard the privacy of individuals or organizations while simultaneously maintaining the integrity and reliability of the data.                                       |

NSL-KDD dataset was created in 2009 by the deletion of superfluous records and the subsequent reorganization of the whole dataset. In order to facilitate the training of the models, the dataset was partitioned into five distinct categories, namely: Denial of Service (DoS), normal, Probe, user to root (U2R), and root to local (R2L) assaults. Nevertheless, the dataset does possess certain limitations, notably the absence of adequate inclusion of low-impact hazards.

#### 5.2.4. Kyoto dataset

The Kyoto dataset, also referred to as Dataset 16, is the subject of discussion. The benchmark data included comprises a total of 24 statistical features, which may be further categorized into 14 conventional features and 10 extra features (Yang et al., 2022). The initial 14 features were derived from the KDD Cup 99 dataset, a widely recognized and extensively utilized dataset for evaluating performance in the field of intrusion detection research. Out of the 41 initial features present in the KDD Cup 99 data set, 14 elements that are deemed significant and vital have been selectively identified and extracted (Hnamte et al., 2023). These features were derived from the raw traffic data collected by honeypot systems deployed at Kyoto University. In addition to the aforementioned 14 features, we have further extracted an additional 10 features that have the potential to enhance our ability to examine network activities with greater efficacy.

#### 5.2.5. CICIDS2017 dataset

The CICIDS2017 dataset is a comprehensive repository designed primarily for analyzing cybersecurity threats within traditional network environments (Abdulghani et al., 2022). The B-profile approach is utilized to generate organic and non-harmful traffic by analyzing human behavior. The dataset encompasses the activities of 25 individuals and includes protocols such as Email, HTTP, HTTPS, SSH, and FTP. Moreover, the study encompasses security attacks on data, including web attacks (e.g., SQL injection, XSS, Brute Force), DDoS attacks (e.g., UDP flood, HTTP flood), and other incidents, including malicious activities within network traffic (Hnamte et al., 2023). While the dataset offers diverse attack scenarios and synthetic benign traffic, its realism might not fully emulate the unique communication patterns and behaviors present in UAV networks that often rely on specialized protocols for command, control, and data transmission, which are not adequately represented in the CICIDS 2017 dataset.

#### 5.2.6. ISCX 2012 dataset

The ISCX 2012 dataset serves as a comprehensive resource for IDS, specifically tailored to network traffic analysis and cybersecurity within enterprise-level environments, providing labeled and anonymized data encompassing various attack scenarios, normal traffic patterns, and anomalies that enable the development, and training (Hnamte et al., 2023; M. et al., 2021). Evaluation of IDS algorithms and models for detecting and mitigating cyber threats within complex network infrastructures. Profiles for agents that generate real traffic for various protocols such as HTTP, SMTP, SSH, IMAP, POP3, and FTP are created by analyzing actual traces. In addition, the ISCX dataset contains a mix of normal and malicious traffic, encompassing different types of attacks, such as DoS, probing, and malware activities. The ISCX 2012 dataset is of a considerable size, allowing for substantial analysis and robust model training while maintaining relevance to real-world network scenarios. Besides that, it includes traffic sessions of varying lengths, providing insights into short-lived and prolonged network interactions.

#### 5.2.7. CSE-CIC-IDS2018 dataset

In order to train prediction models on network-based intrusion detection, datasets like CSE-CIC-IDS2018 were produced (Hnamte et al., 2023). These datasets are intended to support research on anomaly-based detection using different machine learning techniques rather than to act as archives for signature-based detection systems. About 16,000,000 instances were gathered over 10 days for CSE-CIC-IDS2018. It is the most recent big data intrusion detection dataset that is accessible to the public and covers a variety of attack scenarios. The origin of CSE-CIC-IDS2018 was ISCX2012. The progression from ISCX2012 to CICIDS2018 witnessed an expansion in represented protocols, aiming for a more realistic reflection of contemporary network environments by mainly including HTTPS, addressing a substantial portion of real-world network traffic (Momand et al., 2023). There is a class imbalance in this multi-class dataset; around 17% of the instances contain attack (anomalous) traffic. Many studies using CICIDS2018 seemed to overlook the significant class imbalance, which can skew results and affect the reliability of intrusion detection models.

#### 5.2.8. AWID2 dataset

The AWID2 dataset consists of two subsets: a larger set of packets denoted as F, and a smaller set denoted as R. The two versions under consideration are not causally connected, meaning that the smaller version has not been derived or generated from the bigger version (Leandro Marcos da Silva and others, 2022). The specimens were obtained at various points in time, utilizing distinct equipment and within diverse environmental conditions. Each iteration consists of a training set, designated as Trn, and a test set, denoted as Tst. The test version has not been derived from the relevant training set. In conclusion, a revised iteration is presented wherein designations are assigned to distinct attacks (ATK), alongside an alternative version whereby the attack designations are categorized into three primary classes (CLS). In this scenario, the datasets solely exhibit variation in the assigned labels. The AWID3 dataset comprises instances of assaults conducted on an enterprise Wi-Fi network. The existing AWID2 dataset is expanded in the study to include 21 distinct types of attacks. These attacks are categorized into two groups: 801.11-specific attacks and attacks targeting higher layers. The targets of these attacks can be either the local network or external systems. The virtual environment was developed in a laboratory setting to accurately simulate a standard enterprise network infrastructure. It is available in two forms, namely CSV and pcap.

#### 5.2.9. UKM-IDS20 dataset

The dataset under consideration is the UKM-IDS20 dataset (Mohammed et al., 2023). The UKM-IDS20 dataset has been collected and presented as a novel benchmark dataset within the (IDS) field. The dataset has a total of 46 distinct attributes, encompassing four prevalent

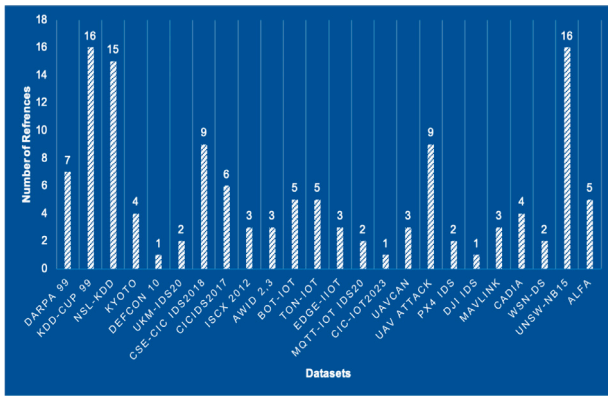


Fig. 13. Datasets against the number of references.

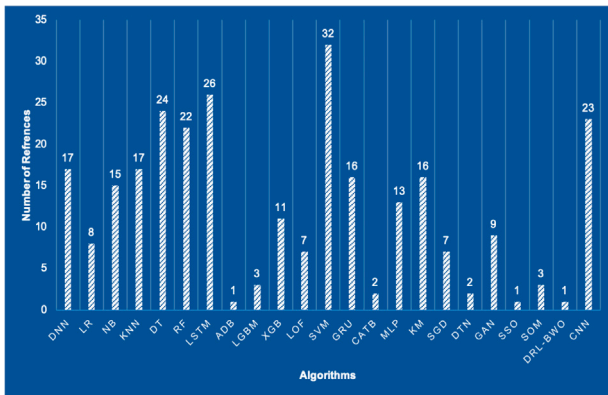


Fig. 14. Algorithms against the number of references.

categories of network attacks, including denial of service. The process of developing the UKM-IDS20 dataset had a duration of two weeks. The establishment of records pertaining to the regular connection occurred during the initial week, whereas the simulation of network attacks utilizing the tools was carried out during the subsequent week. During the second week, the attack operations were recreated on an individual basis. The T-Shark utility was utilized to gather the raw packets from the network.

#### 5.2.10. DEFCON-10 dataset

The dataset referred to as DEFCON-10 dataset, specifically identified as dataset number 18, is the subject of academic interest (Kheddar et al., 2023). The DEFCON-10 dataset was generated in 2002, through the utilization of flawed packets, the FTP over telnet protocol, administrator privileges, port scanning, and sweep attacks. The Capture the Flag (CTF) competition produces a distinct type of traffic compared to real-world network traffic. This distinction arises from the predominance of attack activity in CTF competitions, as opposed to the presence of ordinary background traffic. Consequently, the use of CTF-generated traffic for testing IDS is limited. The dataset is primarily utilized for the purpose of assessing the effectiveness of alert correlation algorithms.

### 5.3. IDS datasets specific for IoT systems

This section delves into a curated collection of datasets tailored explicitly for IDS within IoT environments, featuring prominent datasets such as Bot-IoT, TON-IoT, EDGE-IoT, CIC-2023, and MQTT-IoT. These datasets encapsulate a diverse array of network traffic, device interactions, and security threats unique to IoT ecosystems, providing invaluable resources for evaluating and enhancing IDS capabilities in

safeguarding IoT infrastructures against cyber threats and vulnerabilities.

#### 5.3.1. Bot-IoT dataset

The dataset titled “Bot-IoT”: is being referred to. The Bot-IoT dataset was developed under a realistic network scenario (Kheddar et al., 2023). The environment encompasses both botnet traffic and ordinary traffic. For the purpose of facilitating the process of labeling, the files were partitioned based on the specific threat type and corresponding subcategory (M. et al., 2021). The collection comprises many types of cyberattacks, such as keylogging, DoS, distributed denial-of-service (DDoS), as well as operating system (OS) and service scan attacks. The categorization of DDoS and DoS intrusions is additionally depending on the protocol employed.

#### 5.3.2. ToN-IoT dataset

The concept of Time-Sensitive Networking for the Industrial Internet of Things encompasses the collection of several data sources inside the overall IIoT system. These specificities are considered within ToN-IoT dataset (Taslimasa et al., 2023; M. et al., 2021). These sources include telemetry data obtained from connected devices, records from Linux and Windows operating systems, as well as industrial IoT network traffic (Escorcía-Gutiérrez et al., 2023). A subcategory called “attack type” was included to specify the specific types of attacks observed, including Denial of Service (DoS), Distributed Denial of Service (DDoS), data injection, ransomware, password attacks, backdoor exploits, cross-site scripting (XSS), scanning activities, and Man-in-the-Middle (MITM) attacks. The aforementioned attacks were initiated with the specific intention of targeting a range of Internet of Things (IoT) and industrial IoT devices.

#### 5.3.3. Edge-IIoTset dataset

The Edge-IIoTset dataset: was generated by employing a collection of devices, sensors, protocols, and cloud/edge configurations (Yassen and Mohammed, 2023). The data is generated by a range of IoT devices that detect temperature and humidity, including ultrasonic sensors, among others. Moreover, it has the capability to detect a total of fourteen instances of attacks against IoT communication protocols, which may be categorized into five distinct sorts of threats: (DoS) and (DDoS) attacks, information-gathering attacks, man-in-the-middle attacks, infiltration attacks, and malware assaults. Furthermore, it offers a multitude of features sourced from diverse channels, encompassing alerts, system assets, records, and network communication. Out of the 1176 features that were previously identified and found to have strong connections, an additional 61 features have been identified as potentially useful.

#### 5.3.4. MQTT-IoT-IDS2020 dataset

The MQTT-IoT-IDS2020 dataset is a pioneering effort in simulating a network infrastructure based on the MQTT protocol for IoT applications (Taslimasa et al., 2023). The dataset is derived using a simulated MQTT network architecture. The network consists of twelve sensors, a broker, a simulated camera, and an adversary. The study documents five distinct scenarios: (1) regular operation, (2) aggressive scan, (3) UDP scan, (4) Sparta SSH brute-force, and (5) MQTT brute-force attack. The raw pcap data are stored, followed by the extraction of features. The extraction process involves identifying and categorizing features at three levels of abstraction from the raw pcap files (Hentati and Fourati, 2020). These levels include: (a) packet features, which pertain to individual packets within the network traffic, (b) unidirectional flow features, which focus on the characteristics of one-way flows of network traffic, and (c) bidirectional flow features, which encompass the attributes of two-way flows of network traffic. The dataset contains csv feature files that are suitable for utilization in Machine Learning (ML) applications. Moreover, the unprocessed pcap files are appropriate for conducting an in-depth study of MQTT IoT network communication and the corresponding attacks.

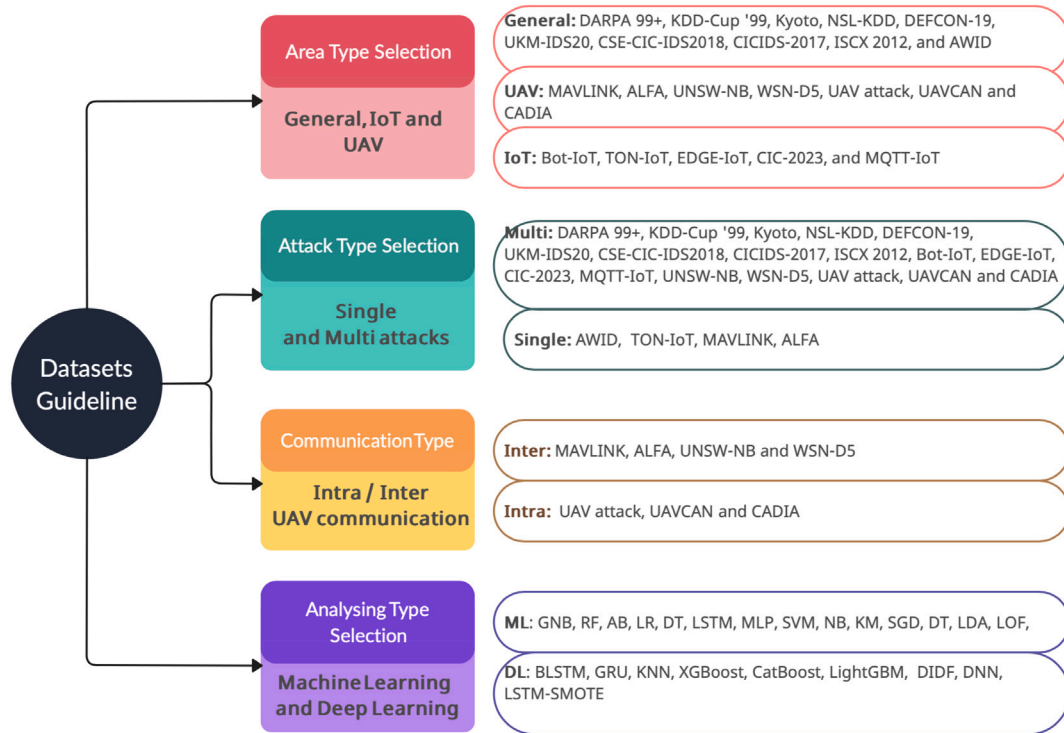


Fig. 15. Proposed Guidelines.

**Table 7**  
Comparative study of datasets for IDS with metrics.

| Ref                                       | Datasets                                  | Techniques               | Accuracy %                         | Precision %                        | Recall %                           | F1-Score %                         |
|-------------------------------------------|-------------------------------------------|--------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Tlili et al. (2023)                       | UA, ALFA, CADIA                           | LSTM, B-LSTM, GRU        | 96.3, 97.3, 96.6                   | N/A                                | N/A                                | N/A                                |
| Mehmood et al. (2022)                     | CSE CIC-IDS 2018                          | SVM, KNN, RF             | 87.5, 90.1, 96.2                   | N/A                                | N/A                                | N/A                                |
| Neto et al. (2023)                        | CIC-IoT2023                               | LR, Pre, AdB, RF, DNN    | 80.2, 81.9, 60.7, 99.1, 98.6       | N/A                                | N/A                                | N/A                                |
| Escorcia-Gutierrez et al. (2023)          | UNSW-NB15, TON-IoT                        | STFA-HDLID               | 99.51, 98.85                       | 99.36, 99.73                       | N/A                                | N/A                                |
| Praveena et al. (2022)                    | NSL-KDD Cup                               | DRL-BWO                  | 98.9                               | 98.5                               | 99.3                               | 98.8                               |
| Ajakwe et al. (2024)                      | UAV Attack                                | OC-SVM, LOF, Autoencoder | N/A                                | N/A                                | N/A                                | 81.17, 58.93, 94.81                |
| Park et al. (2020)                        | MAVLink                                   | SITL, HITL               | N/A                                | N/A                                | N/A                                | 96.01, 96.0                        |
| Shrestha et al. (2021)                    | CSE-CICIDS-2018                           | DT, KNN, LDA, GNB        | 99.99, 99.99, 94.45, 76.15         | N/A                                | N/A                                | N/A                                |
| Silva et al. (2023)                       | ALFA                                      | MLP, SVM, GRB, RF        | 92.0, 96.0, 97.0, 97.0             | N/A                                | N/A                                | N/A                                |
| Abdulghani et al. (2022)                  | CIC-IDS2017                               | LSTM-SMOTE               | 99.83                              | 97.95                              | 99.79                              | 98.96                              |
| Ihekoronye et al. (2022)                  | NSL-KDD                                   | RSCV                     | 97.24                              | 96.54                              | 95.93                              | 96.38                              |
| Morel et al. (2020)                       | MAVLink                                   | LR, NB, KNN, DT, AdB, RF | 94.7, 94.7, 98.8, 98.8, 95.3, 89.4 | 50.0, 50.0, 94.8, 91.7, 55.7, 89.4 | 72.3, 72.3, 92.3, 95.7, 95.9, 96.7 | 59.9, 59.1, 93.5, 93.7, 70.3, 92.9 |
| Park et al. (2021)                        | UAV Attack, ALFA                          | Fault Detection Mode     | 99.69                              | N/A                                | N/A                                | N/A                                |
| He et al. (2023)                          | CIC-IDS2017                               | FCL-SBLS                 | 95.1                               | 94.8                               | 94.3                               | 94.6                               |
| Fadhila et al. (2023)                     | UAVCAN                                    | DIDF                     | 1                                  | 1                                  | 92.0                               | 96.0                               |
| Samriya et al. (2023)                     | KDD Cup 99, NSL-KDD                       | ACO-DNN                  | 98.0, 93.0                         | 98.5, 93.8                         | 99.2, 93.2                         | 99.3, 93.8                         |
| Shar et al. (2022)                        | DJI, ArduPilot, PX4                       | LSTM                     | N/A                                | 96.30                              | 96.80                              | 96.70                              |
| Ferrao et al. (2023)                      | ALFA                                      | LSTM, B-LSTM, GRU        | 84.5, 90.7, 80                     | N/A                                | N/A                                | N/A                                |
| Leandro Marcos da Silva and others (2022) | AWID2                                     | XGB, CatB, LGBM          | 99.81, 99.73, 99.82                | N/A                                | N/A                                | 97.0, 95.0, 95.0                   |
| M. et al. (2021)                          | BoT-IoT, CICIDS2017, ISCX 2012, UNSW-NB15 | KNN                      | 52.9, 70.1, 69.8, 82.6             | 87.3, 86.4, 93.7, 95.4             | 40.4, 53.2, 52.6, 66.7             | 55.3, 65.8, 67.4, 78.5             |
| Alissa et al. (2022)                      | KDD-Cup, NSL-KDD                          | CSODAE-ID                | 99.12                              | 98.25                              | 98.24                              | 98.24                              |
| Jullian et al. (2022)                     | TEXBAT dataset                            | MLP, LSTM                | 83.23, 82.1                        | 87.07, 91.04                       | 67.14, 75.58                       | 82.79, 82.59                       |
|                                           | MAVLINK dataset                           | MLP, LSTM                | 99.93, 99.93                       | 99.96, 100                         | 85.71, 85.71                       | 92.29, 92.31                       |



**Table 8**

Comparative study of datasets for IDS Intra/Inter UAV communication.

| Ref                                       | Datasets                                     | Year | Targeted Domain | ML | DL | Techniques                         | Attacks                                        | Category | Intra/Inter |
|-------------------------------------------|----------------------------------------------|------|-----------------|----|----|------------------------------------|------------------------------------------------|----------|-------------|
| Whelan et al. (2020)                      | UAV Attack dataset                           | 2020 | UAV             | Y  | Y  | OC-SVM, LOF                        | Malicious                                      | Private  | Intra       |
| Ajakwe et al. (2024)                      | UAV Attack dataset                           | 2022 | UAV             | Y  | Y  | OC-SVM, LOF, OC-AE                 | Malicious                                      | Private  | Intra       |
| Shar et al. (2022)                        | (DJI, ArduPilot, PX4) datasets               | 2022 | UAV             | Y  | Y  | LSTM                               | (vel, com), (acc, gyro), (acc, gyro, GPS, mag) | Private  | Intra       |
| Fadhila et al. (2023)                     | UAVCAN dataset                               | 2023 | UAV             | Y  | Y  | DIDF                               | Flooding, Replay, Fuzzing                      | Public   | Intra       |
| Tlili et al. (2023)                       | UA, ALFA, CAIDA datasets                     | 2023 | UAV             | Y  | Y  | LSTM, BLSTM, GRU                   | spoofing, jamming, DoS                         | Public   | Intra       |
| Abdulghani et al. (2022)                  | CICIDS2017 dataset                           | 2017 | General         | Y  | Y  | LSTM-SMOTE                         | Null                                           | Public   | Inter       |
| Morel et al. (2020)                       | BOUN DDoS dataset                            | 2020 | General         | Y  | Y  | LR, NB, DT, RF, etc                | DDoS                                           | Public   | Inter       |
| Shrestha et al. (2021)                    | CSE-CIC-IDS2018 dataset                      | 2021 | General         | Y  | Y  | KM, SGD, DT, KNN, LDA, and LR      | Botnet, DoS, Web, BF, DDoS, Infiltration       | Public   | Inter       |
| M. et al. (2021)                          | ISCX 2012, BoT-IoT, CICIDS2017 and UNSW-NB15 | 2021 | General         | Y  | Y  | k-Nearest Neighbor                 | HTTP DoS, DDoS, SSH Brute Force                | Public   | Inter       |
| Ihekoronye et al. (2022)                  | NSL-KDD dataset                              | 2022 | General         | Y  | Y  | RSCV                               | Probe, DoS, U2R, R2L                           | Public   | Inter       |
| Leandro Marcos da Silva and others (2022) | AWID2 dataset                                | 2022 | General         | Y  | Y  | XGBoost, CatBoost, LightGBM        | DoS                                            | Public   | Inter       |
| Mehmood et al. (2022)                     | CSE-CIC-IDS2018 dataset                      | 2022 | General         | Y  | Y  | RF,SVM, KNN                        | Null                                           | Public   | Inter       |
| Praveena et al. (2022)                    | NSL-KDD Cup dataset                          | 2022 | General         | Y  | Y  | Black Widow Optimization           | Probe, DoS, U2R, R2L                           | Public   | Inter       |
| Halisdemir et al. (2024)                  | NSL, UNSW-NB15 dataset                       | 2022 | General         | N  | N  | Annual Locked Shields              | DoS                                            | Public   | Inter       |
| Mohammed et al. (2023)                    | UKMIDS20 dataset                             | 2022 | General         | Y  | Y  | FS-MLP                             | ARP poisoning, DoS, Scans, Exploit             | Public   | Inter       |
| Samriya et al. (2023)                     | KDD Cup 99, NSL-KDD datasets                 | 2023 | General         | Y  | Y  | ACO-DNN                            | Null                                           | Public   | Inter       |
| Hnamte et al. (2023)                      | CICIDS2017, CICIDS2018                       | 2023 | General         | Y  | Y  | LSTM-AE                            | DoS                                            | Public   | Inter       |
| He et al. (2023)                          | CIC-IDS2017 dataset                          | 2023 | General         | Y  | Y  | (FCL-SBLS) based on (DTN)          | Bot, BF, DoS, PortScan, Web Attack             | Public   | Inter       |
| Silva et al. (2023)                       | WSN-DS dataset                               | 2023 | General         | Y  | Y  | LightGBM, XGBoost, MLP, NB, RF, DT | blackhole, gray-hole, flooding                 | Public   | Inter       |
| Morel et al. (2020)                       | NWPU-VHR10 dataset                           | 2020 | IoT             | Y  | N  | LR, NB, KNN, DT, AdB, RF           | DoS, UDP flood, Replay MITM                    | Public   | Inter       |
| Yassen and Mohammed (2023)                | Edge-IIoTset datasets                        | 2023 | IoT             | Y  | Y  | GNB K-NN, RF, AB, LR, DT, LSTM     | DDoS, SQL inj, vul scanner, backdoor           | Public   | Inter       |
| Neto et al. (2023)                        | CICIoT2023 dataset                           | 2023 | IoT             | Y  | Y  | LR, Perceptron, and Adaboos        | Backdoor, DDoS, DoS, MITM, etc                 | Public   | Inter       |
| Escorcia-Gutierrez et al. (2023)          | TON-IoT, UNSW-NB15 datasets                  | 2023 | IoT             | Y  | Y  | Sparrow Search Optimization        | Backdoor, DDoS, DoS, MITM, etc                 | Public   | Inter       |
| Park et al. (2020)                        | MAVILNK dataset                              | 2020 | UAV             | Y  | Y  | DNN                                | DoS, GPS Spoofing                              | Public   | Inter       |
| Park et al. (2021)                        | UAV Attack dataset, ALFA dataset             | 2021 | UAV             | Y  | N  | fault detection                    | DoS                                            | Private  | Inter       |
| Jullian et al. (2022)                     | MAVLINK dataset                              | 2022 | UAV             | Y  | Y  | performance                        | GPS spoofing                                   | Public   | Inter       |
| Ferrao et al. (2023)                      | ALFA dataset                                 | 2023 | UAV             | Y  | Y  | MLP, SVM, Gradient boosting, RF    | Malicious                                      | Public   | Inter       |

### 5.3.5. CICIOT2023 dataset

The CICIOT2023 Dataset introduces an innovative dataset of IoT attacks specifically designed for security analytics applications in practical IoT operations (Neto et al., 2023). In an IoT topology consisting of 105 devices, it carries out a total of 33 attacks. These attacks are categorized into seven types: DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, and Mirai. The dataset comprises IoT devices functioning as both aggressors and targets. The study also gathers data from 33 cyber-attacks and assesses the effectiveness of machine and deep learning algorithms in accurately categorizing and identifying whether IoT network traffic is hostile or benign.

## 6. Guidelines and lessons learned toward datasets selection for IDS within UAVs communication systems

Choosing an appropriate dataset for UAV security research can be time-consuming due to various factors. Here, our aim is to assist researchers by providing a set of recommendations and scenarios specifically tailored to the UAV context. This means that the dataset must accurately represent real-life scenarios that researchers seek to replicate or investigate, ensuring that the findings are relevant and applicable to practical situations. To optimize the selection of datasets for this purpose, we conducted a comparative study, analyzing the differences and similarities between the datasets under consideration. Figs. 13 and 14 respectively illustrate the comparison of these datasets in relation to the number of references and the algorithms employed in the studies. These visual comparisons provide insights into how extensively the datasets have been referenced and which algorithms have been applied, helping to identify the most suitable dataset for specific research objectives. In addition to the visual comparisons, Tables 7 and 8 offers a detailed analysis of the research articles that have cited the use of these datasets. These tables summarize the frequency of references, the contexts in which the datasets were used,

and the effectiveness of the datasets in contributing to the research outcomes. This comprehensive analysis helps to understand the relevance and impact of each dataset within the research community, guiding researchers in selecting the most appropriate datasets for their investigations specifically in Intra/Inter. These recommendations serve as a guide for selecting the most suitable dataset based on contextual parameters relevant to UAV security as shown in Fig. 15. To aid researchers in intelligent UAV studies, we focus on several crucial metrics. Primarily, it is vital to ensure that the dataset aligns with the current study's issues or problems. Besides that, the data can be analyzed based on its information level, collection technique, presentation approach, and features aimed at mitigating and addressing the imbalance of objective correlation in data sets related to the penetration of wireless network-connected devices. Particularly the specific file format that encompasses this data. The efficacy of these devices and their extent of integration inside the network. In this context, the proposed guideline to assist researchers in selecting Intrusion Detection System (IDS) datasets specifically tailored for UAVs' communication, encompassing both intra and inter-UAV communications takes into consideration several aspects:

- **Datasets Accessibility:** Researchers have access to several datasets located in online data repositories, which can be either publicly or privately accessible. The disparity related to the data access is an important issue when selecting the dataset.
- **Contextual Relevance:** Ensure the dataset aligns with UAV communication scenarios, considering the unique aspects of both intra-UAV (within a single UAV) and inter-UAV (between multiple UAVs) communication environments. In these study, we categorized IDS datasets into generic datasets and datasets specifically relevant to the IoT and UAVs. Therefore this study helps researchers in using the relevant datasets for UAV communication.

- **Documentation and Metadata:** In general, datasets that come with comprehensive documentation, detailing the data collection methodology, characteristics of the communication instances, and any potential biases or limitations are more useful for performing research studies and allow researchers to compare their studies to other works.
- **Realism, Diversity, and Data Authenticity:** Datasets could be synthetic or based on real traffic collected from real-world scenarios. In general, researchers should look for datasets that emulate diverse real-world communication scenarios encountered by UAVs, including variations in environmental conditions, terrain, signal interference, and potential threats. Besides that, datasets that consist of authentic, actual communication data or carefully simulated data that accurately represent the challenges and complexities of UAV communication systems should be prioritized.
- **Dataset Size and Variability:** Assess the dataset's size and variability to ensure it contains a sufficient volume and diversity of communication instances, covering various communication protocols, frequencies, and potential intrusion scenarios.
- **Anomaly and Intrusion Types:** Seek datasets that encompass a wide range of anomalies and potential intrusions relevant to UAV communication, including signal jamming, spoofing, message tampering, and other security threats. By considering these factors, researchers can make informed decisions when selecting IDS datasets tailored for UAV communication security, enhancing the development and evaluation of effective intrusion detection systems for UAV networks.

## 7. Conclusion

UAV network security datasets serve as fundamental resources aiding in the detection of potential attacks and the fortification of security measures. These datasets are instrumental for researchers aiming to craft robust security protocols, ensuring the reliability and safety of UAV networks. This study offers an extensive examination of IDS datasets dedicated to assess attacks within UAVs communication systems. It commences by providing an overview of the UAV network environment, and prevalent security concerns. Furthermore, it introduces a novel classification system for existing UAV network IDS datasets, categorizing them into three primary groups: Generic use IDS datasets, IoT-based IDS datasets, and UAVs-based IDS datasets. Besides that the datasets dedicated to handle UAVs communications have been subcategorized into three subcategories: (1) IDS datasets supporting intra-UAVs communications, (2) IDS datasets supporting inter-UAVs communications between UAV and GCS, and (3) IDS datasets supporting inter-UAVs communications between multiple UAVs. Additionally, it outlines the salient attributes of UAV networks as depicted in the available datasets. Finally, a comprehensive comparative analysis of diverse datasets is presented to aid UAV network researchers in selecting the most suitable dataset tailored to their specific requirements.

## CRedit authorship contribution statement

**Ahmad Burhan Mohammed:** Writing – review & editing, Writing – original draft, Visualization. **Lamia Chaari Fourati:** Writing – review & editing, Writing – original draft, Methodology, Investigation.

## Intellectual property

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No data was used for the research described in the article.

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## Further reading

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